Experiment 8: Snort an IDS to detect malicious activities

IT Network Security Lab

Course Code: **CSSF3124**

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Jan – June, 2025

LAB EXERCISE 8

Objectives:

(1) Establish Network of two Systems using VirtualBox

(2) Install IDS and detect various Network activities

Installing Snort (IDS / IPS)

https://www.snort.org

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What is Snort?

Snort is the foremost Open Source Intrusion Prevention System (IPS) in the world. Snort IPS uses a series of rules that help define malicious network activity and uses those rules to find packets that match against them and generates alerts for users.

Snort can be deployed inline to stop these packets, as well. Snort has three primary uses: As a packet sniffer like tcpdump, as a packet logger — which is useful for network traffic debugging, or it can be used as a full-blown network intrusion prevention system. Snort can be downloaded and configured for personal and business use alike.

Introduction to Snort

- Initial Snort released as IDS in 1998
- Snort 2.x started supporting as IPS, 2021
- Snort 3.x further improved over Snort 2.x released in 2023
- **Snort 2.x was widely used.** It being an open source project, has very good community support for doucmentation and rule-sets.

Introduction to Snort

How to run: https://docs.snort.org/start/help

How to create Rules: https://docs.snort.org/start/rules

Download Community Rules: https://www.snort.org/downloads/#rule-downloads

Types of Rule sets for Snort

Types of Rules:

1) User defined (Local)

2) By Talos (have to Pay)

3) Community Rules (Free)

Installing Snort

Configuring FTP Server

Use the command:

sudo apt-get update

sudo apt-get install snort

1

Verify IP range (mine): 192.168.29.0/24

Supply the range when installation prompts:

192.168.29.0/24

2

```
upes@upes-Latitude-5420:~$ ip a s
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN gro
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s31f6: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_cod
    link/ether 08:92:04:07:73:fc brd ff:ff:ff:ff:ff:ff
3: wlp113s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue s
    link/ether c8:94:02:1c:3[illegible]1f brd ff:ff:ff:ff:ff:ff
    inet 192.168.29.189/24 brd 192.168.29.255 scope global dynamic nopr
       valid_lft 77386sec preferred_lft 77386sec
    inet6 2405:201:6804:42c9:257:3d5e:6663:6e89/64 scope global tempora
       valid_lft 7329sec preferred_lft 7329sec
    inet6 2405:201:6804:42c9:2a71:60e5:3245:8ba6/64 scope global dynami
       valid_lft 7329sec preferred_lft 7329sec
    inet6 fe80::4e52:66f9:2916:c310/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
upes@upes-Latitude-5420:~$
```

Note in my Oracle VirtualBox I have installed a machine with IP Set in the same range:

192.168.29.0/24

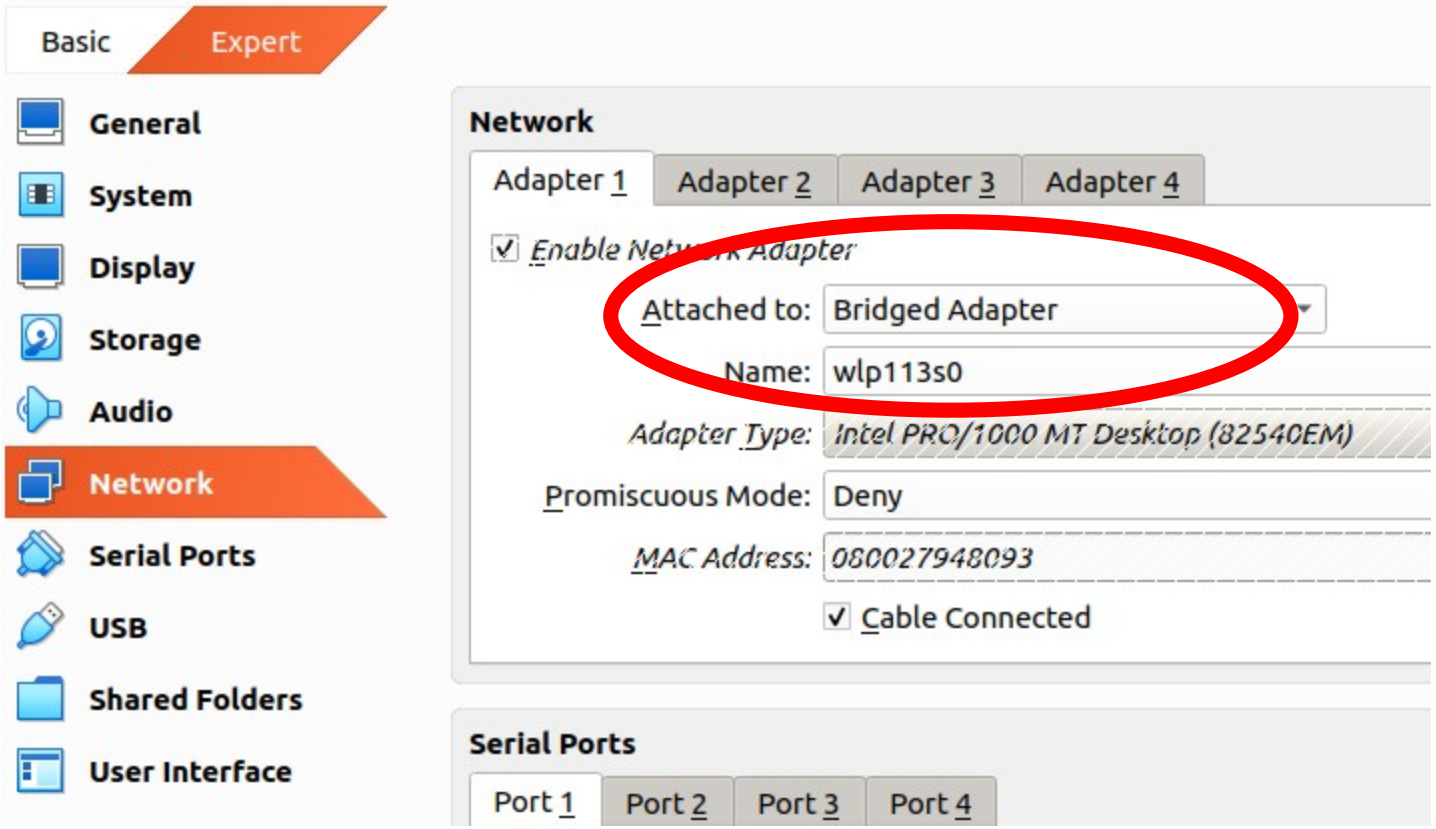
VirtualBox/VMWare Network Connection Type

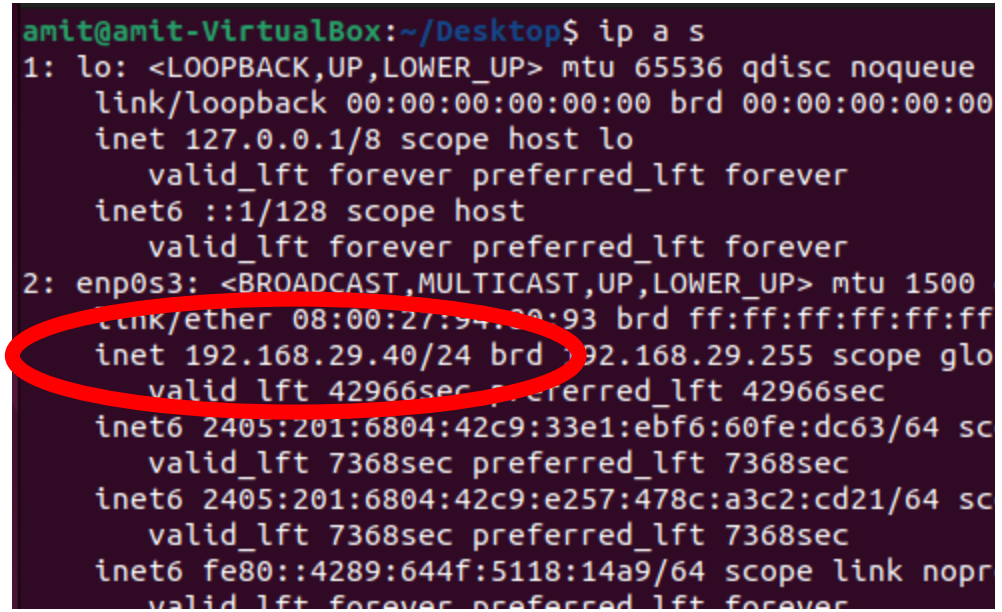
Communication

	VM ↔ VM	VM → Host	VM ← Host	VM → LAN	VM ← LAN
Not attached	–	–	–	–	–
NAT	+	+	Port Forward	+	Port Forward
NAT Network	+	+	Port Forward	+	Port Forward
Bridged	+	+	+	+	+
Internal Network	+	–	–	–	–
Host-only	+	+	+	–	–

In Hindi: https://www.youtube.com/watch?v=hL0lgwyMroU&t=827s

VirtualBox Network Setting



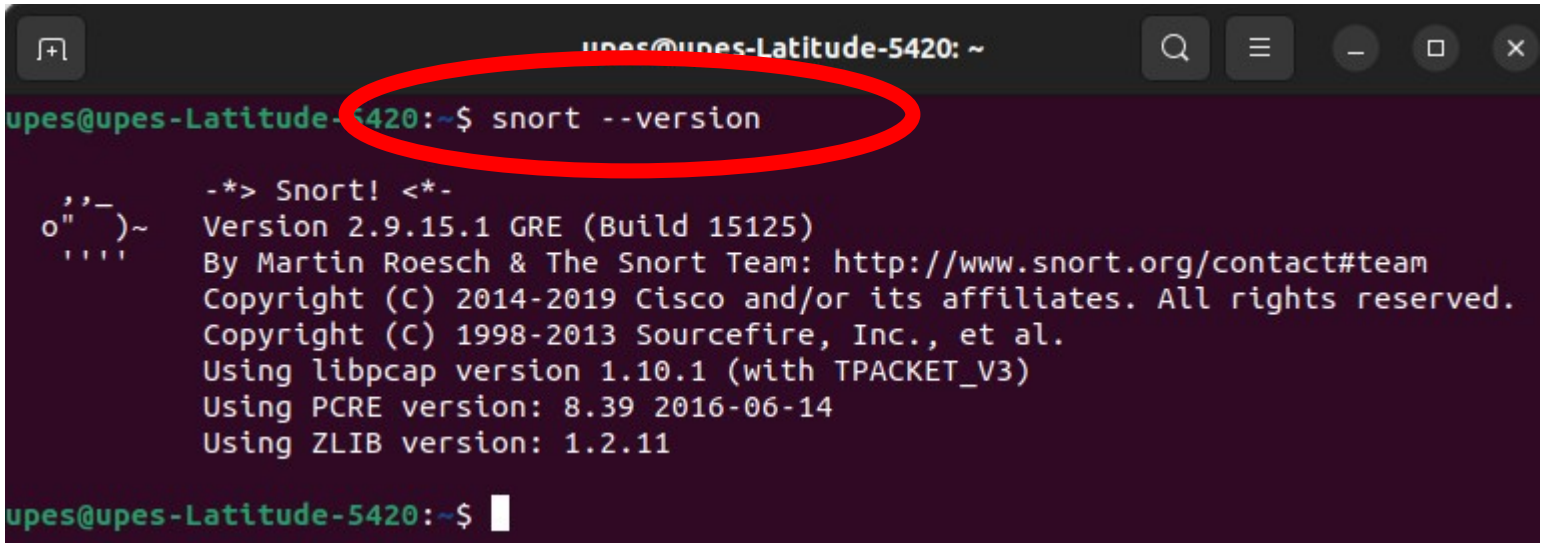


Note in my Oracle VirtualBox I have installed a machine with IP Set in the same range: 192.168.29.0/24

Understanding Snort Configurations

Test installation:

snort --version **1**



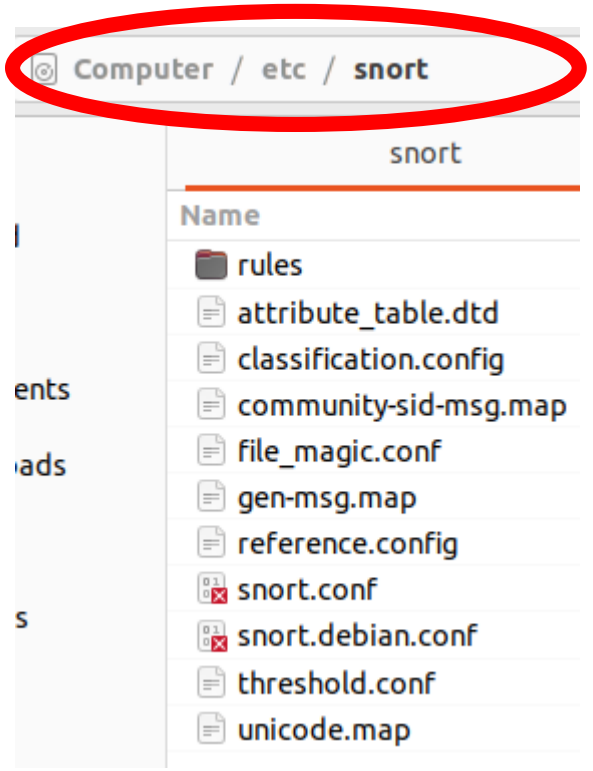
To allow Snort listen to all trafic, set the Promiscuous Mode to ON (No restrictions policy).

For VirtualBox: you will find it in Network Setting

For host, use the command:

sudo ip link set **wlp113s0** **promisc** **on**

Understanding Snort Configurations



Local.rules: this is the user/custom rules set by the local user.

icmp.rules: this is the standard rules set provided by the community.

Understanding Snort Configurations

Recommendation: Create a copy of the file "/etc/snort/snort.conf" before any modification

sudo cp /etc/snort/snort.conf /etc/snort/snort_original.conf

Now open the configuration file: **/etc/snort/snort.conf**

From Line 25 – 37: This section describes the format of the different sections of this configuration file.

We, need to do the following setup:

Step-1) Network variables (HOME_NET): **ipvar HOME_NET 192.168.29.0/24**

Also verify the Rule_Path if set properly as: RULE_PATH /etc/snort/rules

Step-7) Check the Rule section: **verify the included rules**

include $RULE_PATH/local.rules

This is the file where we will create our rules.

Note that community rules that are available in the folder **/etc/snort/rules/** are also included in the file.

Test Snort Configurations

To verify if the configuration file satifies the syntax:

sudo snort -T -i wlp113s0 -c /etc/snort/snort.conf

Use

-T: start up in self-test mode,

-c use the rules located in the config-file

Use: man snort to find the help on using different options in Snort

Setting our First Rule for Snort for ICMP Detection

Open the local rule file

Type at the end:

1

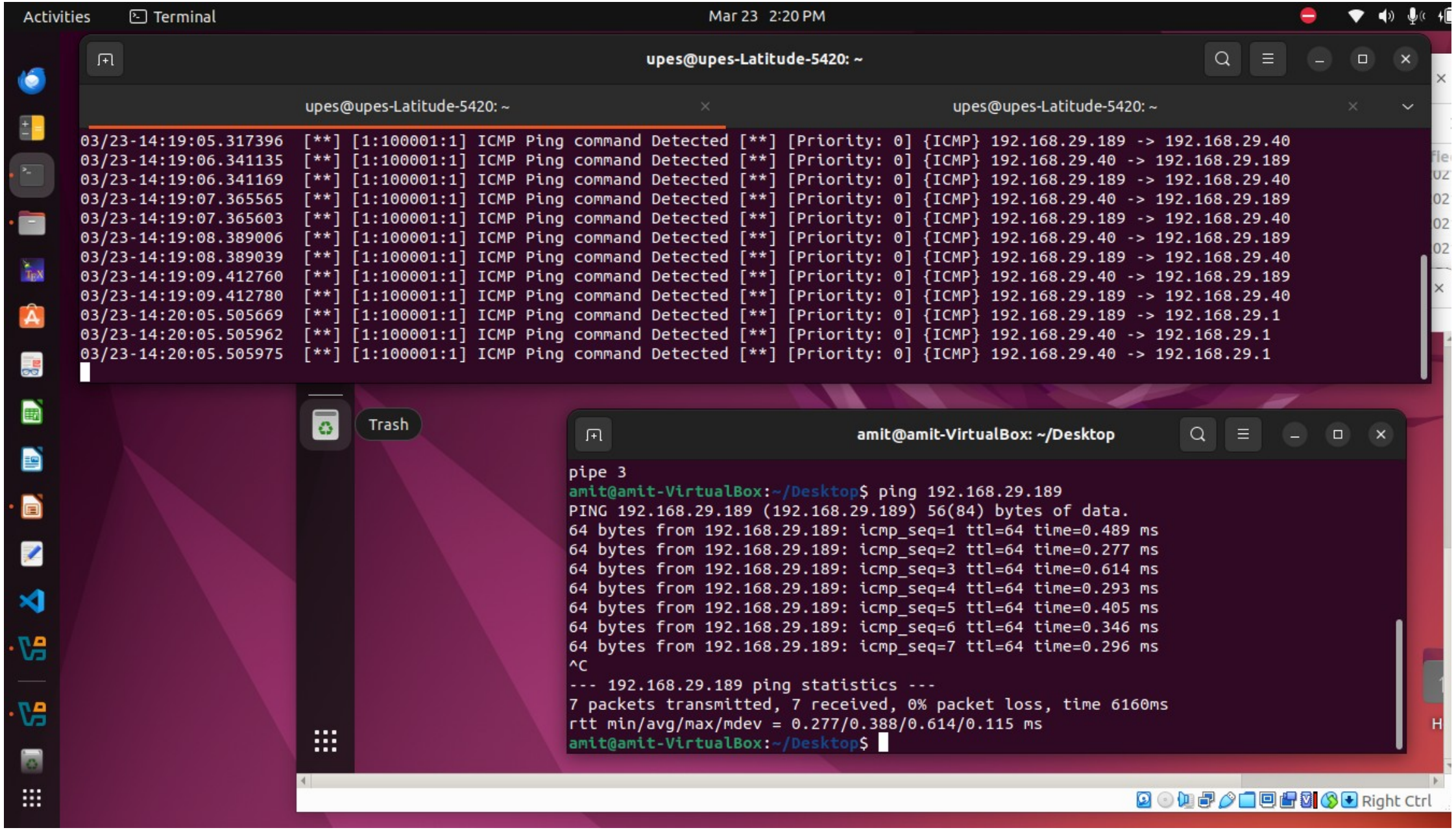
```
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping command Detected"; sid:100001; rev:1;)
```

To Run Snort type:

2

```
sudo snort   -q    -l  /var/log/snort  -i  wlp113s0  -A console  -c /etc/snort/snort.conf
```

Snort Detecting Ping Activity



Setting Rule for FTP Login Detection

Open the local rule file

Type at the end:

1

```
alert tcp any any -> $HOME_NET 21 (msg:"FTP Connection request Detected"; sid:100001; rev:1;)
```

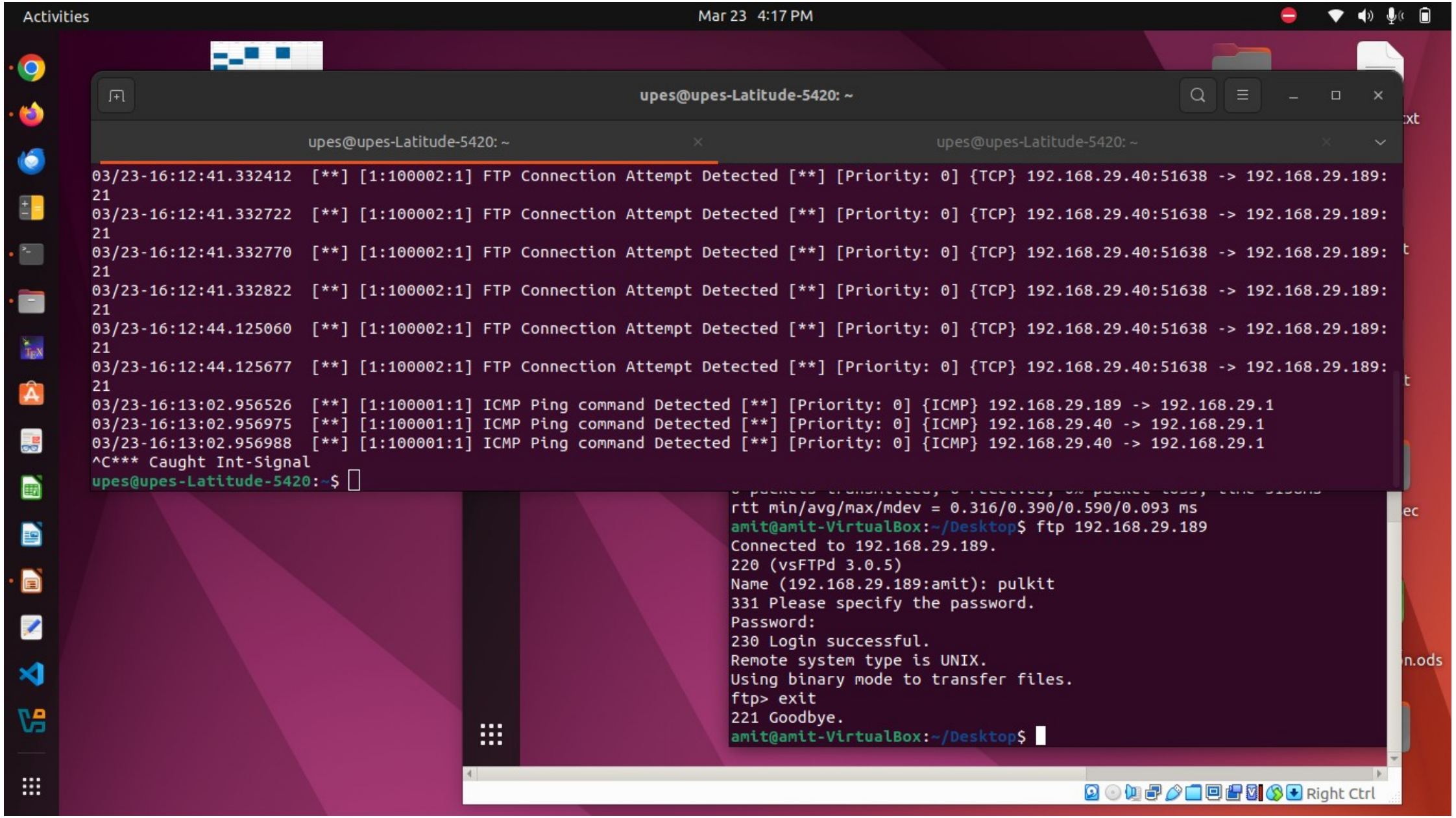
FTP uses reliable TCP connection using port 21

2

To Run Snort type:

```
sudo snort   -q    -l  /var/log/snort  -i  wlp113s0  -A console  -c /etc/snort/snort.conf
```

Snort Detecting FTP Activity



Setting Rule on Content

Open the local rule file

Type at the end:

1

alert tcp any any -> $HOME_NET 80 (msg:"Content Rule – 1: Hacked Detected"; content:"Hacked"; nocase; sid:100004; rev:1;)

alert tcp any any -> $HOME_NET 4444 (msg:"Hack Detected in Netcat Chatting"; content:"Hack"; nocase; sid:100005; rev:1;)

Content Rule detecting on port 80, 4444

2

To Run Snort type:

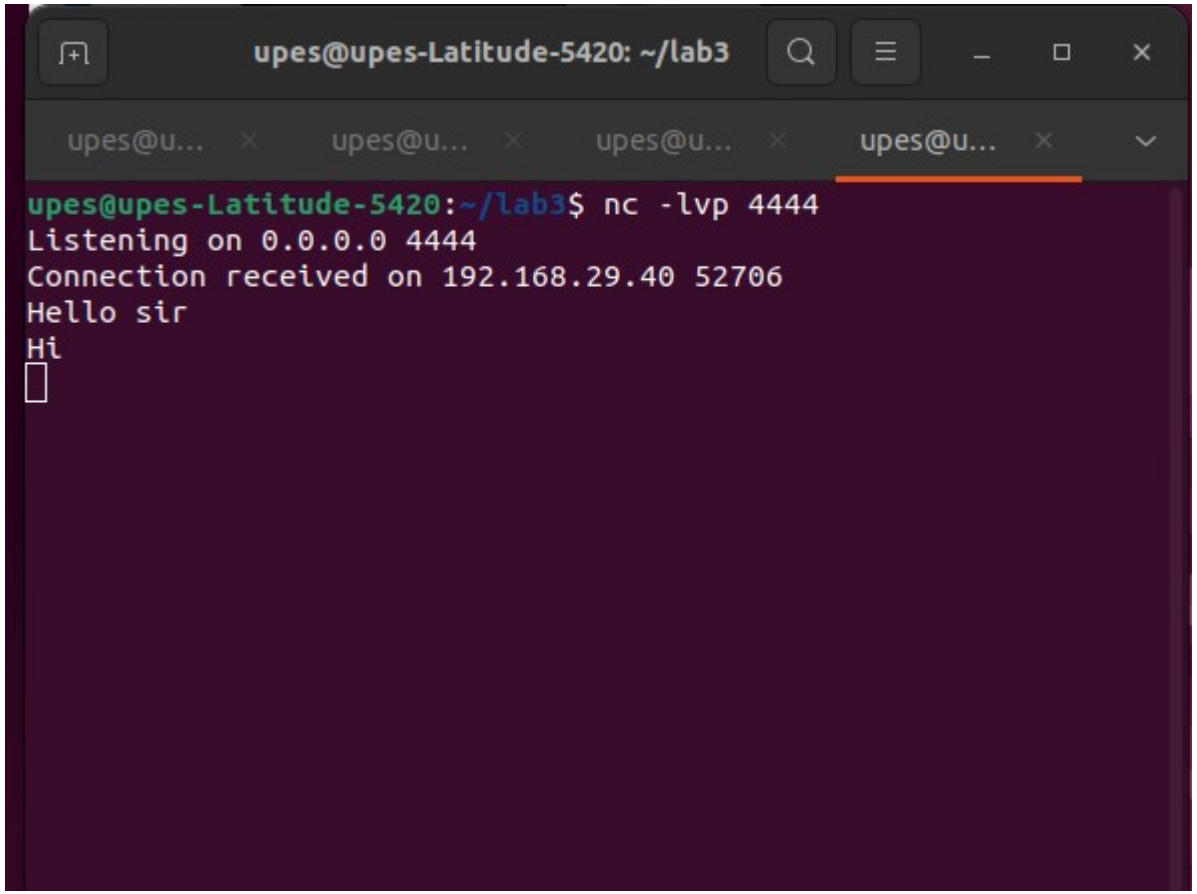
sudo snort -q -l /var/log/snort -i wlp113s0 -A console -c /etc/snort/snort.conf

Starting a Simple NetCat Chatting

Open Terminal and type:

1

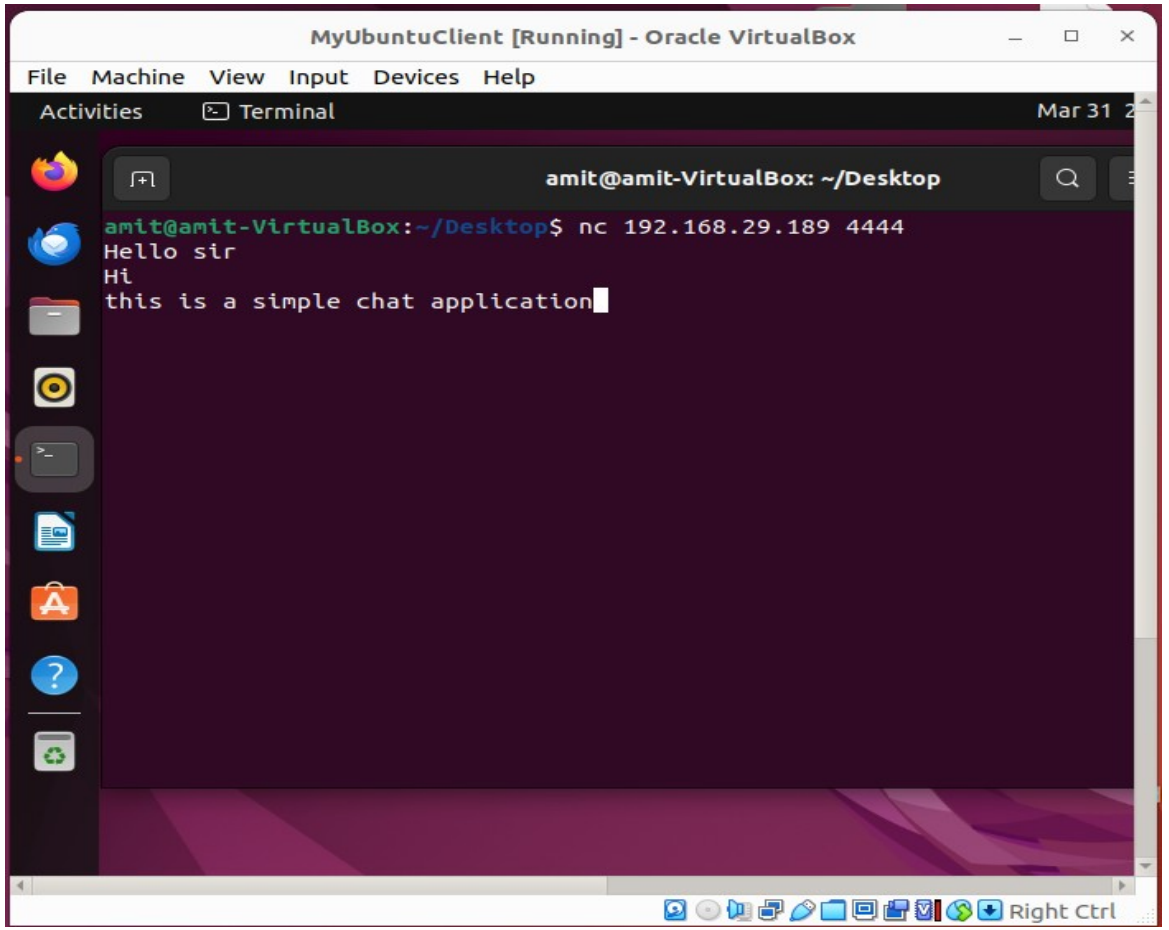
nc -lvp <server_ip> <port_no>



Open another terminal or connect from remote system:

2

nc <server_ip> <port_no>



Assignment

Assignment will be uploaded in the LMS